**3. Develop an Android application implementing contextual menus and popup menus.**

**ProjectName: MenuDemo**

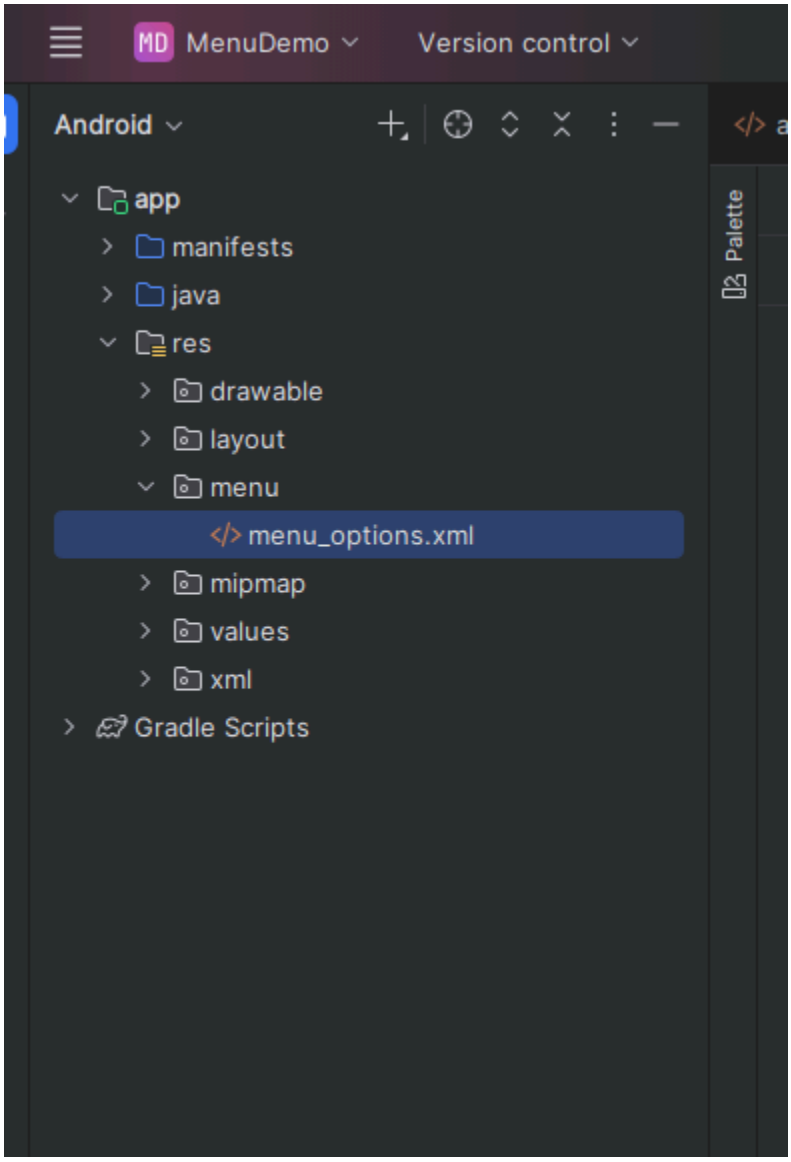
**File Name: activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <TextView
       android:id="@+id/txtContext"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Long Press Me"
       android:textSize="24sp"
       android:padding="20dp"/>
   <Button
       android:id="@+id/btnPopup"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Show Popup Menu"
       android:layout_marginTop="30dp"/>
</LinearLayout>
```

**Create a Menu Resource File.**
**Right Click: res→ New → Android Resource Directory**
**Directory Name: menu**
**Resource Type: menu**
**Click OK**
Right Click: menu → New → Menu Resource File

**File Name : menu_options**



**File Name : menu_options.xml**

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">

   <item

       android:id="@+id/menu_edit" android:title="Edit"/>

   <item

       android:id="@+id/menu_delete"

       android:title="Delete"/>

   <item

       android:id="@+id/menu_share"

       android:title="Share"/>

</menu>
```

**File Name: MainActivity.java**

```
package com.example.menudemo;

import android.os.Bundle;

import android.view.ContextMenu;

import android.view.MenuItem;

import android.view.View;

import android.widget.Button;

import android.widget.PopupMenu;

import android.widget.TextView;

import android.widget.Toast;

import androidx.annotation.NonNull;

import androidx.appcompat.app.AppCompatActivity;
```

```java
public class MainActivity extends AppCompatActivity {


   TextView txtContext;

   Button btnPopup;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       txtContext = findViewById(R.id.txtContext);

       btnPopup = findViewById(R.id.btnPopup);


       // Register Context Menu

       registerForContextMenu(txtContext);


       // Popup Menu

       btnPopup.setOnClickListener(v -> {


           PopupMenu popupMenu = new PopupMenu(MainActivity.this, btnPopup);


           popupMenu.getMenuInflater().inflate(

                   R.menu.menu_options,

                   popupMenu.getMenu());
```

```
        popupMenu.setOnMenuItemClickListener(item -> {
            Toast.makeText(MainActivity.this,
                    item.getTitle(),
                    Toast.LENGTH_SHORT).show();
            return true;
        });

        popupMenu.show();
    });
}

// Context Menu Creation
@Override
public void onCreateContextMenu(ContextMenu menu,
                                View v,
                                ContextMenu.ContextMenuInfo menuInfo) {

    super.onCreateContextMenu(menu, v, menuInfo);

    menu.setHeaderTitle("Select Option");
    getMenuInflater().inflate(R.menu.menu_options, menu);
}

// Context Menu Click Handling
@Override
public boolean onContextItemSelected(@NonNull MenuItem item) {
```

```
        Toast.makeText(this,
                item.getTitle(),
                Toast.LENGTH_SHORT).show();


        return true;
    }
}
```

